Do Now · retrieval

Lesson 110 · Round Robin and Multilevel Feedback Queues

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which scheduler preempts a running task when a shorter one arrives?

Q2

Why can an array provide direct indexed access?

Q3

A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

Q4

For the binary value 1010, what is its decimal value?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Round robin — Ready processes receive a time quantum in cyclic order.

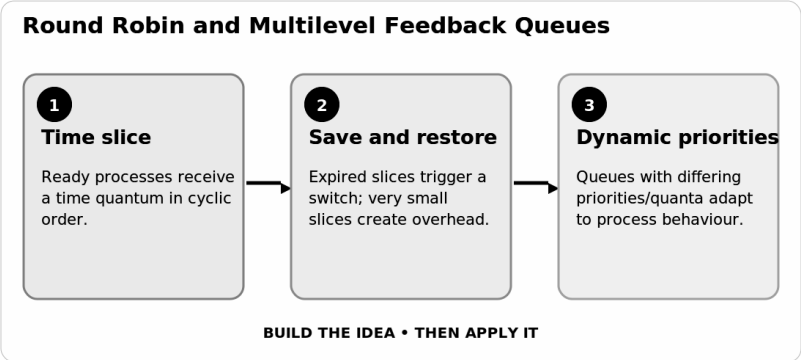
Core idea 2

Context switching — Expired slices trigger a switch; very small slices create overhead.

Core idea 3

MLFQ — Queues with differing priorities/quanta adapt to process behaviour.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

time quantum — Maximum continuous CPU interval per RR turn.

round robin — Preemptive cyclic scheduling.

context switch — Saving/restoring CPU state when changing process.

MLFQ — Multilevel feedback queue scheduling.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

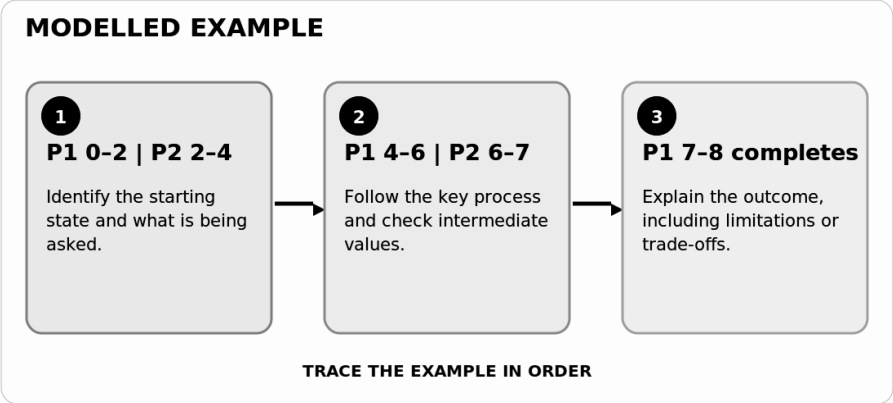
Problem

RR with quantum 2: P1 burst 5, P2 burst 3; both arrive at t=0. Trace completion.

Follow the worked method

1. Model answer — P1 runs 0–2, P2 2–4, P1 4–6, P2 6–7 (finishes), P1 7–8 (finishes). Completion times: P2 7, P1 8. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Describe the round-robin algorithm, including the purpose of the time quantum. [6]

Question 2

RR quantum 2: A burst 3, B burst 4, both arrive 0. Produce a Gantt timeline. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain how MLFQ responds to CPU-bound and interactive tasks and evaluate a very short quantum. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Reducing a time quantum to an extremely small value always improves performance.

Use key vocabulary from memory

1. Explain time quantum in your own words.

2. Explain round robin in your own words.

3. Explain context switch in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.